

STA 35C: Statistical Data Science III

Lecture 18: Multiple Hypotheses Testing

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Announcement

Homework 5 is due tomorrow

Midterm 2 on Fri, May 15 (1:10 pm–2:00 pm in class)

- **Arrive early:** The exam starts at 1:10 pm and ends at 2:00 pm sharp
- **One hand-written cheat sheet:** Letter-size (8.5"×11"), double-sided, brief formulas/notes
- **Calculator:** A simple (non-graphing) scientific calculator is allowed
- **No other materials** beyond the single cheat sheet (no textbooks, etc.)
- **SDC accommodations:** Confirm scheduling with AES online ASAP

Advice for preparation:

- Primary coverage: Lectures 12–19
- Core concepts from earlier may be assumed, especially regression and model assessment
- Two practice midterms and brief solution keys are posted on course webpage
- Office hours next week:
 - Instructor: Wed, 3:30–4:30 pm
 - TA: Tue 12:30–2:30pm

Agenda

Recap: Challenges & motivation for multiple hypotheses testing

- Issues arising with multiple tests
- Real-world concerns: p-hacking & data dredging

Today: How to control false positives

- **Family-wise error rate (FWER)**
 - Definition & intuition
 - Controlling FWER: Bonferroni correction & Holm's step-down
- **False discovery rate (FDR)**
 - Definition & intuition
 - Controlling FDR: Benjamini–Hochberg procedure

Recap: Multiple testing

Single-hypothesis test:

- Typically set up H_0 , and gather data to reject it if there is significant evidence
- Type I error = false positive; Type II error = false negative
- Each test has Type I error at most α (e.g. 0.05)

Modern data analysis: multiple tests simultaneously

- E.g. Testing thousands of predictors or biomarkers to discovery significant ones
- If m is large, false rejections can occur easily by chance even when all nulls are true
- If all m nulls are true and each test is at level α , the expected number of false positives is $m \times \alpha$

Key challenge: Address and adjust for the inflation of false positives as m grows

Articles warning about misused statistical significance



Figure: Many reproducibility crises trace back to undisclosed multiple testing or selective reporting. Proper adjustments can help mitigate these issues.

Related issues: p -hacking and data dredging

Real danger: Searching for “significant” results in many ways until something “works”

- Repeatedly testing different hypotheses/subgroups
- Eventually, some test may yield $p < 0.05$ *by chance*
- Example: trying many outcomes, subgroups, transformations, or model specifications, then reporting only the smallest p -value

Outcomes: Spurious “discoveries”

- Published claims may fail to replicate
- True findings can be overshadowed by noise

Needs: Systematic multiple-testing corrections are crucial, especially for large m

Recall: Hypothesis testing as classification

A single hypothesis test makes a decision about whether to reject H_0 :

- **Goal:** Discover a "real phenomenon" (H_1) or conclude not enough evidence (H_0)
 - H_0 is true $\iff$ no real effect
 - H_0 is false $\iff$ there is a real effect (H_1)
 - We claim a "discovery" when H_0 is rejected
- **Analogy to classification:** Depending on the evidence in the data,
 - Reject $H_0 \iff$ classify $\hat{H} = 1$
 - Fail to reject $H_0 \iff$ classify $\hat{H} = 0$

	H_0 is true (" H=0 ")	H_0 is not true (" H=1 ")
Reject H_0 (" $\hat{H}=1$ ")	Type-I error (FP)	Correct (TP)
Not reject H_0 (" $\hat{H}=0$ ")	Correct (TN)	Type-II error (FN)

$\implies \Pr(\text{Type I error}) = \Pr(\text{reject a true null}) \approx \frac{FP}{FP+TN}$ empirically among true nulls

- By setting a threshold α , we want to control $\Pr(\text{Type I error})$ below α

Family-wise error rate (FWER): Definition

Single test: $\Pr(\text{Type I error}) = \Pr(\text{reject a true null})$

Multiple tests (m hypotheses): Familywise error rate (FWER) is defined as

$$\begin{aligned}\text{FWER} &= \Pr(\text{reject at least one true } H_0) \\ &= \Pr(\# \text{ Type-I error} \geq 1),\end{aligned}$$

i.e. the probability of *any* false positive among m tests

If all m null hypotheses are true, the tests are independent, and each are at level α :

$$\text{FWER} = 1 - (1 - \alpha)^m,$$

- When $m = 1$, $\text{FWER} = 1 - (1 - \alpha)^m = 1 - (1 - \alpha) = \alpha$
- Grows quickly with m
 - E.g. $m = 20$, $\alpha = 0.05 \implies \text{FWER} \approx 0.64 \gg 0.05$

Visualization: FWER grows as m increases

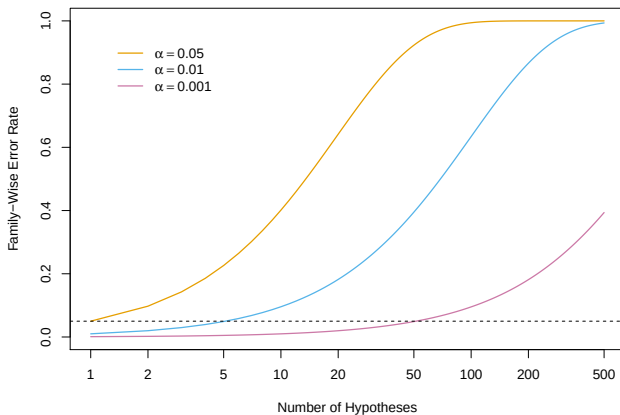


Figure: FWER vs. number of tests m (log scale) for $\alpha = 0.05$ (orange), 0.01 (blue), 0.001 (purple). The dashed line is 0.05. For $m = 50$ and target FWER=0.05, each test must be at $\alpha = 0.001$ [JWHT21, Figure 13.2].

Testing multiple hypotheses at level α

Suppose we test m hypotheses $H_{0,1}, \dots, H_{0,m}$, all at level α , obtaining confusion matrix:

	H_0 is true	H_0 is not true
Reject H_0	N_{FP}	N_{TP}
Not reject H_0	N_{TN}	N_{FN}

- $N_{\text{FP}}, N_{\text{TP}}, N_{\text{TN}}, N_{\text{FN}}$ are random variables that sum to m
- Roughly, we expect $N_{\text{FP}} \approx m \cdot p_{\text{FP}}$; when all m nulls are true, $N_{\text{FP}} \approx m \cdot \alpha$

If these m tests are independent,

- Probability of *at least one* false positive $\approx 1 - (1 - \alpha)^m$
- For $m = 20, \alpha = 0.05$, that probability is $\approx 64\%$

Goal of FWER control: Keep the probability of making even one false positive small:

$$\text{FWER} = \Pr(N_{\text{FP}} \geq 1) \leq \alpha$$

The Bonferroni correction for FWER control

Key idea: Use the union bound over the true null hypotheses:

$$\text{FWER} = \Pr \left(\sum_{j=1}^m \{\text{Reject } H_j\} \right) \leq \sum_{j=1}^m \Pr (\{\text{Reject } H_j\})$$

- Each test is done at level $\alpha/m \implies \Pr (\{\text{Reject } H_j\}) \leq \alpha/m \implies \text{FWER} \leq \alpha$

The Bonferroni method (Bonferroni correction):

- For each hypothesis $H_1, \dots, H_m$, **reject H_j if $p_j < \frac{\alpha}{m}$** (instead of $< \alpha$)

Pros & Cons:

- **Pros:** Simple & widely used
- **Cons:** Often *very conservative* $\implies$ few rejections (=discoveries) & lower power¹

¹Power = TPR = the fraction of false null hypotheses that are successfully rejected

Example: Bonferroni correction

Example

Let $m = 6$ hypotheses with p-values:

$$p_1 = 0.0018, \quad p_2 = 0.009, \quad p_3 = 0.021, \quad p_4 = 0.034, \quad p_5 = 0.045, \quad p_6 = 0.070.$$

At $\alpha = 0.05$, threshold $= \frac{\alpha}{m} = \frac{0.05}{6} \approx 0.00833$.

Reject H_j if $p_j < 0.00833$.

Hence:

$$p_1 = 0.0018 < 0.00833 \implies \text{reject } H_1,$$

but $p_2 = 0.009 > 0.00833$ and the rest are larger. So Bonferroni rejects only H_1 .

Conclusion: 1 rejection using Bonferroni, whereas naive $p < 0.05$ would reject 5 of them $(p_1, \dots, p_5)$.

Holm's step-down procedure for FWER control

Holm's method refines Bonferroni to be less conservative:

Holm's method

- 1 Specify α , the level at which to control the FWER
- 2 Compute the p -values for the m null hypotheses, $H_{01}, \dots, H_{0m}$
- 3 Sort p -values so that $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$
- 4 Define

$$L = \min \left\{ j : p_{(j)} > \frac{\alpha}{m + 1 - j} \right\}$$

- 5 Reject all null hypotheses H_{0j} for which $p_j < p_{(L)}$

Properties:

- Ensures $\text{FWER} \leq \alpha$
- Rejects at least as many hypotheses as Bonferroni

Example: Holm's step-down procedure

Example

Step 1: Set $\alpha = 0.05$

Step 2: $p_1 = 0.0018, p_2 = 0.009, p_3 = 0.021, p_4 = 0.034, p_5 = 0.045, p_6 = 0.070$.

Step 3: Sort p -values $p_{(1)} = 0.0018, p_{(2)} = 0.009, p_{(3)} = 0.021, p_{(4)} = 0.034, p_{(5)} = 0.045, p_{(6)} = 0.070$.

Step 4: Find $L = 3$ because

$$p_{(1)} = 0.0018 ? 0.0018 \leq \frac{0.05}{6 + 1 - 1} = \frac{0.05}{6} \approx 0.00833 \quad \Rightarrow \text{reject } H_{(1)}, \text{ continue}$$

$$p_{(2)} = 0.009 ? 0.009 \leq \frac{0.05}{6 + 1 - 2} = \frac{0.05}{5} = 0.01 \quad \Rightarrow \text{reject } H_{(2)}, \text{ continue}$$

$$p_{(3)} = 0.021 ? 0.021 \leq \frac{0.05}{6 + 1 - 3} = \frac{0.05}{4} = 0.0125? \text{ No} \quad \Rightarrow \text{stop; } L = 3$$

Step 5: We reject $H_{(1)}, H_{(2)}$ total 2 rejections. The rest are not rejected.

Conclusion: Holm's method rejects 2, whereas Bonferroni rejected only 1.

Visualization: Bonferroni vs. Holm

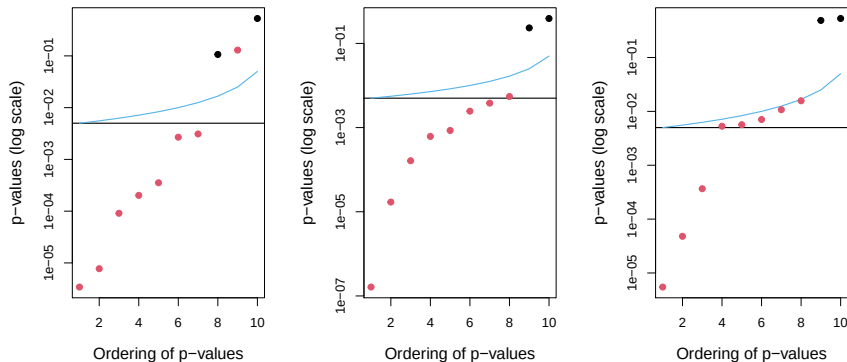


Figure: Each panel shows sorted p -values from a separate simulation of $m = 10$ null hypotheses, with the two true nulls in black and the others in red. Controlling the FWER at 0.05, Bonferroni rejects all points below the **black** line, while Holm rejects all below the **blue** line. The gap between these lines indicates the additional hypotheses Holm rejects but Bonferroni does not. In the middle panel, Holm rejects one more null than Bonferroni; in the right panel, it rejects five more [JWHT21, Figure 13.3].

Pop-up quiz: Bonferroni vs. Holm

Suppose $m = 4$, $\alpha = 0.05$, and the sorted p -values are

$$p_{(1)} = 0.010, \quad p_{(2)} = 0.015, \quad p_{(3)} = 0.030, \quad p_{(4)} = 0.200.$$

Question: How many hypotheses are rejected by Bonferroni and by Holm's method?

- A) Bonferroni rejects 1; Holm rejects 2.
- B) Bonferroni rejects 2; Holm rejects 1.
- C) Both reject 1.
- D) Both reject 2.

Answer: A. Bonferroni uses $0.05/4 = 0.0125$, so it rejects only $p_{(1)}$. Holm compares $0.010 \leq 0.0125$, then $0.015 \leq 0.05/3$, then stops at $0.030 > 0.05/2$, so it rejects two.

Illustration: Power (TPR) vs. FWER trade-off

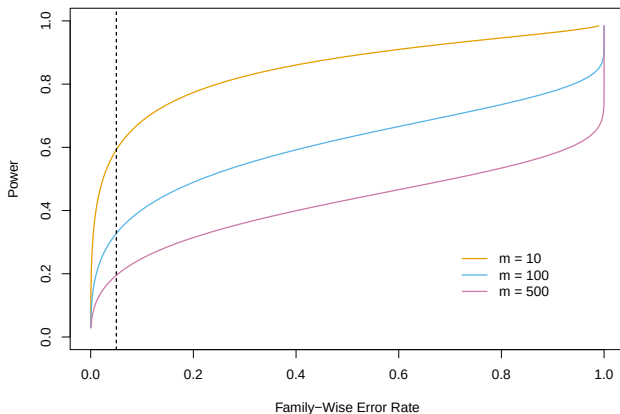


Figure: In a simulation with 90% of m nulls true, the power is displayed against FWER. Colors of the curves: $m = 10$ (orange), $m = 100$ (blue), $m = 500$ (purple). Larger m reduces power. The vertical dashed line marks FWER=0.05 [JWHT21, Figure 13.5].

FWER control may not suffice

FWER demands *no* false rejections (at all) with probability at least $1 - \alpha$:

- Very stringent if m is large
- Tends to reduce power (fewer true positives found)

In modern “exploratory” studies:

- We may tolerate a small fraction of false positives to discover more true ones
- This leads to the *false discovery rate (FDR)* approach

	H_0 is true	H_0 is not true
Reject H_0	Type-I error (FP)	Correct (TP)
Not reject H_0	Correct (TN)	Type-II error (FN)

New goal: Find more true discoveries while controlling the proportion of false discoveries.

False discovery rate (FDR): Definition and motivation

Motivation: Controlling FWER can be too conservative for large m

Instead: Control the fraction of rejected hypotheses that are *false positives*

$$\text{FDP} = \frac{\# \text{ false positives}}{\# \text{ total rejections}} = \frac{\# \text{ FP}}{\# \text{ FP} + \# \text{ TP}}$$

- Directly controlling FDP is impossible because we never know which H_{0j} are true/false
- Thus, we aim to control FDP in expectation

False discovery rate (FDR) = $\mathbb{E}[\text{FDP}]$

- Allow up to fraction q of false positives *on average* among the “claimed discoveries”
- The choice of q is context- and dataset-dependent (no gold standard like $\alpha = 0.05$)

Properties:

- Accept a small fraction of false positives, in exchange for more total discoveries
- Typically yields more rejections (“discoveries”) than FWER-based methods

Controlling FDR: Benjamini–Hochberg procedure

Benamini-Hochberg procedure

- 1 Specify q , the level at which to control the FDR
- 2 Compute the p -values for the m null hypotheses, $H_{01}, \dots, H_{0m}$
- 3 Sort p -values so that $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$

4 Define

$$L = \max \left\{ j : p_{(j)} < \frac{qj}{m} \right\}$$

- 5 Reject all null hypotheses H_{0j} for which $p_j \leq p_{(L)}$

Result:

- Under standard conditions, ensures $\text{FDR} \leq q$, but not necessarily small FWER
- Typically more powerful, yielding more rejections, than Bonferroni/Holm if m is large

Example: Benjamini–Hochberg procedure

Example

Step 1: Set $q = 0.05$

Step 2: $p_1 = 0.0018, p_2 = 0.009, p_3 = 0.021, p_4 = 0.034, p_5 = 0.045, p_6 = 0.070$.

Step 3: Sort p -values $p_{(1)} = 0.0018, p_{(2)} = 0.009, p_{(3)} = 0.021, p_{(4)} = 0.034, p_{(5)} = 0.045, p_{(6)} = 0.070$.

Step 4: Find $L = \max \{j : p_{(j)} < \frac{qj}{m}\} = 3$ because

$$\begin{array}{ll} j = 1 : & 0.0018 \leq 0.05 \times \frac{1}{6} \approx 0.0083? \checkmark \\ j = 2 : & 0.009 \leq 0.05 \times \frac{2}{6} \approx 0.0167? \checkmark \\ j = 3 : & 0.021 \leq 0.05 \times \frac{3}{6} = 0.025? \checkmark \\ j = 4 : & 0.034 \leq 0.05 \times \frac{4}{6} \approx 0.0333? \text{ No} \\ j = 5 : & 0.045 \leq 0.05 \times \frac{5}{6} \approx 0.0417? \text{ No} \\ j = 6 : & 0.070 \leq 0.05 \times \frac{6}{6} = 0.05? \text{ No} \end{array}$$

Step 5: Reject $H_{(1)}, H_{(2)}, H_{(3)}$.

Conclusion: BH rejects 3, while Holm rejects 2, Bonferroni rejects 1.

Visual comparison: Bonferroni vs. Benjamini–Hochberg

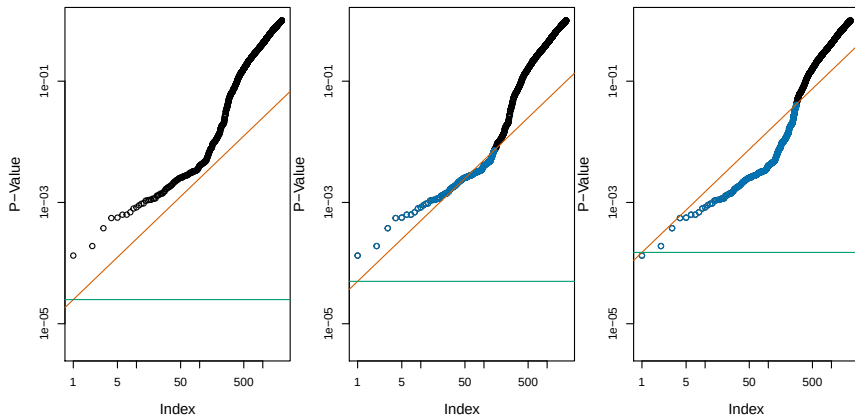


Figure: Panels: same set of $m = 2000$ sorted p-values for the **Fund** dataset. **Green lines:** thresholds for FWER control (Bonferroni) at $\alpha = 0.05, 0.1, 0.3$ (left to right). **Orange lines:** thresholds for FDR control (Benjamini–Hochberg) at $q = 0.05, 0.1, 0.3$ (left to right). E.g., When the FDR is controlled at $q = 0.1$, 146 nulls are rejected (center, blue points). At $q = 0.3$, 279 nulls are rejected (right, blue points) [JWHT21, Figure 13.6].

Pop-up quiz: Comparing FDR vs. FWER

Suppose there are m hypotheses to test simultaneously. **Which statement best captures the difference between FDR and FWER?**

- A) FDR forces the probability of zero false positives to stay below α , whereas FWER allows a small fraction q .
- B) FDR aims to keep $\mathbb{E}[\text{fraction of false positives among rejections}] \leq q$, while FWER demands $\Pr(\text{at least one false positive}) \leq \alpha$.
- C) Under FDR control, no false positives are allowed once you discover enough true positives.
- D) FDR only works for independent tests, but FWER can handle correlated tests without adjustments.

Answer: B.

FDR control (e.g., Benjamini–Hochberg) allows a certain fraction of false positives *on average*, whereas FWER control (e.g., Bonferroni/Holm) requires the chance of *any* false positive be controlled below α .

Wrap-up

- Multiple testing increases the chance of false positives; ordinary $p < 0.05$ rules are not enough when many hypotheses are tested.
- **FWER** controls the probability of making false positives:
 - Strictly ensures $\Pr(N_{\text{FP}} \geq 1) \leq \alpha$, e.g., via Bonferroni/Holm
 - Conservative for large m , leading to fewer rejections & reduced power
- **FDR** controls the expected fraction of false positives among rejections:
 - Controls $\text{FDR} = \mathbb{E}[\text{FDP}] \leq q$, e.g., via Benjamini–Hochberg
 - Typically yields more rejections than FWER, especially when m is large
- **Practical consideration:**
 - Use FWER for strict confirmatory analyses needing minimal Type I error
 - Use FDR for exploratory, large-scale studies, tolerating some false positives to gain more discoveries

References



Gareth James, Daniela Witten, Trevor Hastie, and Robert Tibshirani.

An Introduction to Statistical Learning: with Applications in R, volume 112 of *Springer Texts in Statistics*.

Springer, New York, NY, 2nd edition, 2021.